

Introduction

The *Thesis Archiving and Research Publication Repository System* is a web-based platform designed to centrally store, manage, and provide controlled access to engineering theses and research publications. The system allows authorized personnel to upload and approve documents before they become visible to users. Instead of allowing downloads, the system provides **view-only access** to protect intellectual property while still enabling knowledge sharing. Advanced search and filtering features enable users to quickly locate theses based on engineering discipline, field of research, authorship, year, and keywords.

Users and Key Roles

- **System Administrator**
 - Manages users, roles, and permissions
 - Oversees system configuration and security
 - Monitors system activity and logs
- **Thesis Uploader (Department Secretary)**
 - Uploads thesis documents and metadata
 - Assigns department, program, and field of research
 - Submits thesis for approval
- **Thesis Approver (Faculty / Program Chair / Research Coordinator)**
 - Reviews uploaded theses
 - Approves or rejects submissions
 - Ensures compliance with academic standards
- **Faculty Members**
 - View and search theses

- Filter papers by year of publication, engineering program, and field of study
 - Full-text viewing access is available
 - Downloading of soft copies is not permitted
 - **Students**
 - Search and view thesis papers
 - Filter papers by year of publication, engineering program, and field of study
 - Full-text viewing access is available
 - Downloading of soft copies is not permitted
 - **Guest Users**
 - View basic thesis information (title, abstract, keywords)
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General Functions

- Role-based login and authentication
- Uploading and approval workflow
- View-only document access (PDF viewer)
- Metadata management (title, authors, adviser, year, abstract, keywords)
- Search and filtering by:
 - Engineering program (CPE, EE, ME, CE, etc.)
 - Field of research (e.g., Medical Engineering, AI, Robotics, Renewable Energy)
 - Year of Publication
- Activity logging and audit trail

System Features

- Advanced search bar with auto-suggestions
- Multi-level filtering (program + research field + year)
- Embedded document viewer with disabled download option
- Approval status tracking (Pending, Approved, Rejected)
- Dashboard for uploaders and approvers
- Secure storage of thesis files
- Tagging and categorization system

Performance Requirements

- Search results returned within 2 seconds
- Support at least 100 concurrent users
- Fast loading of PDF viewer
- Secure access control and encryption
- Scalable storage for increasing number of theses
- System uptime of at least 99%